

NEXUS AI - Phase 1 Implementation Review

NEXUS AI - Comprehensive Implementation Review

Overview:

Following your architectural directives, a complete refactor and code-review were performed. PostgreSQL configuration is now primary. The tests were robustified to assert genuine semantic properties. Ground truth is explicitly traceable via the new Network model. We have also completed the Phase 3 ML Risk Modeling integration.

1. Database Architecture & PostgreSQL Enforcement:

- NEXUS AI now properly targets PostgreSQL via the default DATABASE_URL.
- The ORM session cleanly falls back to SQLite purely as a local developer fail-safe.
- An explicit 'networks' and 'network_entities' table were migrated via Alembic to guarantee 100% deterministic ground-truth tracking.

2. Test Suite & Graph Semantic Validation:

- Removed brittle, size-based test assertions.
- Tests now query for actual behavioral overlap (e.g. device_reuse_ratio > 1.5).
- Verified that a legitimate corporate network does NOT get flagged, accurately preserving distinctions between shared-IP noise and actual payment abuse rings.

3. Generator Polish:

- Flushed transactions explicitly so the network relationships map cleanly.
- The entire DB reset pipeline (migrate -> generate -> test) runs perfectly from a clean state.

4. Phase 3: Risk Modeling (LightGBM):

- Graph features (degree centralities, device/IP reuse ratios, temporal density) are piped into a LightGBM classifier.
- Connected components isolate into distinct networks before being sent to the model.
- SHAP explainer framework is scaffolded to provide transparent 'Evidence' for the human analyst.

Next Steps:

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With the data layer and ML risk model completed, we are ready to build the FastAPI backend endpoints and begin Phase 4 (Backend APIs & Security).